

263-5354-00L
Large Language Models

Multimodal Language Models

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Lecture Outline

Vision Language Models

- The general architecture of VLMs
- Case studies of popular VLMs (VisualBERT, CLIP)

Is Language Modelling Everything?

We have seen the rise of large language models...

What is missing from these models trained **solely on text**, even when those corpora are internet-scale?

Humans rely heavily on non-linguistic cues for learning:

- Perception
- Social context
- Spoken language
- External factual and commonsense knowledge

Multimodal Models

It would be useful to build models that combine language with other modalities of information.

Multimodal language models: Models trained on text + one or more other modalities (**images**, speech, audio, knowledge, action, ...).

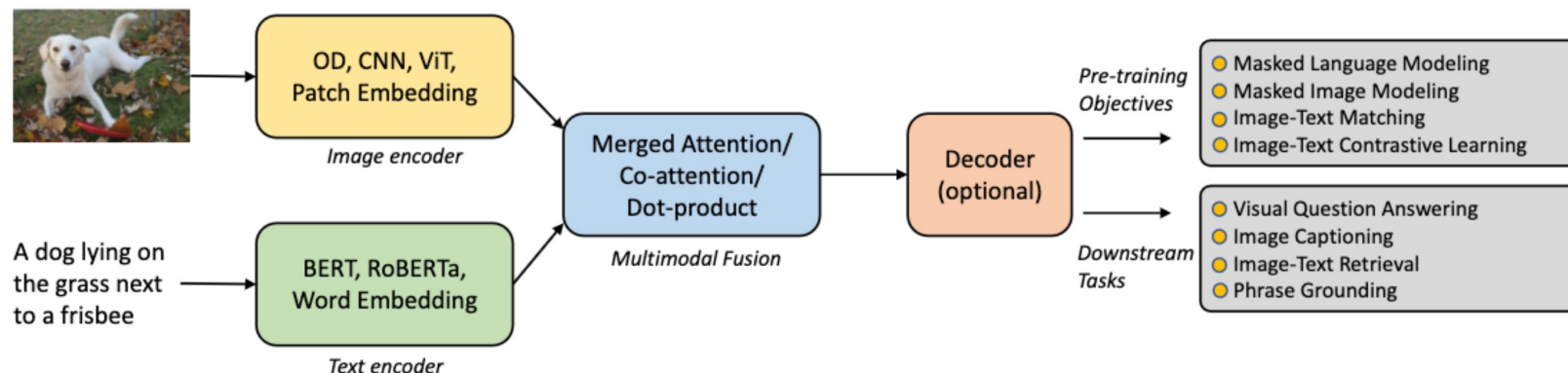
Benefits of multimodal models:

- **Faithfulness** to human experience: Human experience is multimodal
- **Practical**: The internet is multimodal. Many applications require multimodality
- **Data efficiency and availability**:
 - **Efficiency**: Multimodal data is rich and “high bandwidth”
 - **Scaling**: More data is better, and we’re running out of high-quality text data.

Multimodality is a key frontier in the foundation model revolution.

Vision Language Models

Vision Language Models



Components of a vision language model:

1. Text encoder:

extracts **text features** $w = \{w_1, \dots, w_N\}$, $N = \text{num}(\text{text tokens})$

2. Image encoder:

extracts **visual features** $v = \{v_1, \dots, v_M\}$

$M = (\text{visual features})$ e.g., num. image regions/grids/patches

3. Multimodal fusion module

produces cross-modal representations

4. (Optionally) decoder

Often no clear boundaries
between image/text
backbones, multimodal fusion
module, and the decoder

Image Encoders

Key idea: Map an input image into a set of feature vectors $v = \{v_1, \dots, v_M\}$

a) Object detectors

- R-CNN, Faster R-CNN

b) Convolutional neural networks

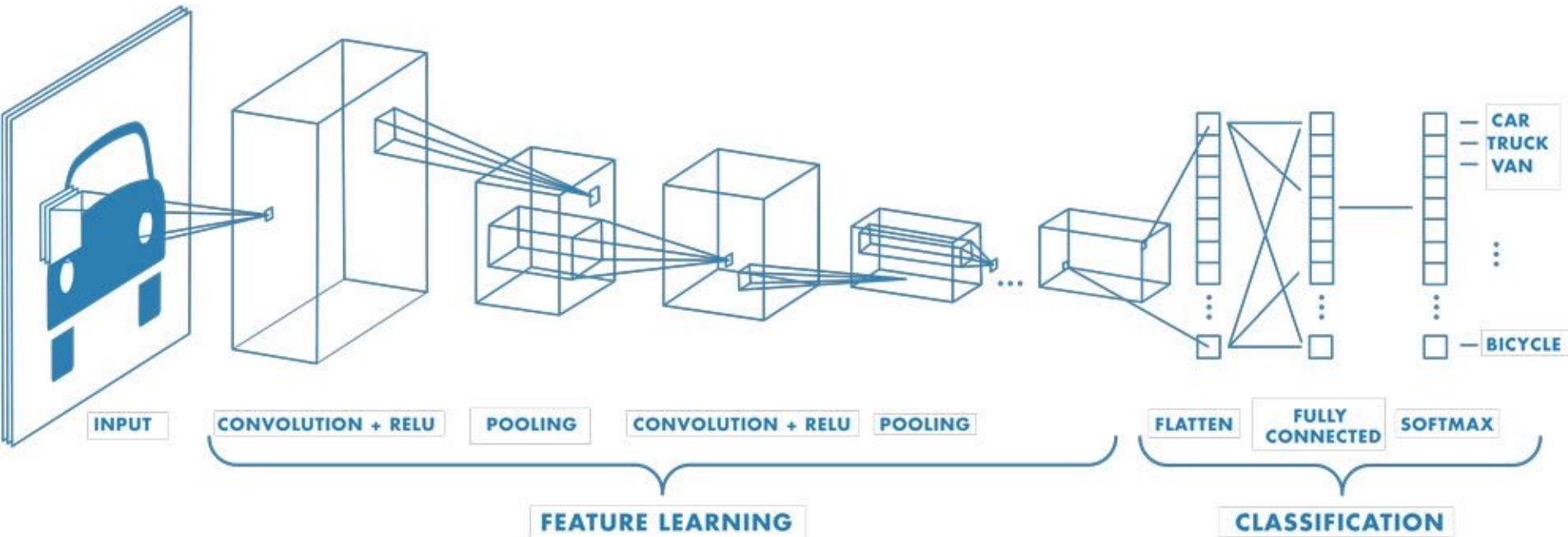
- ResNet pretrained from ImageNet

c) Vision transformers

a) Object detectors

- Start with a pretrained Object detector e.g. R-CNN or Faster R-CNN
- Use visual features as well as location features $[x_1, y_1, x_2, y_2, w, h, w \cdot h]$ (normalized coordinates, width, height and area)
- Both visual and location features are combined, e.g., fed through a fully-connected layer

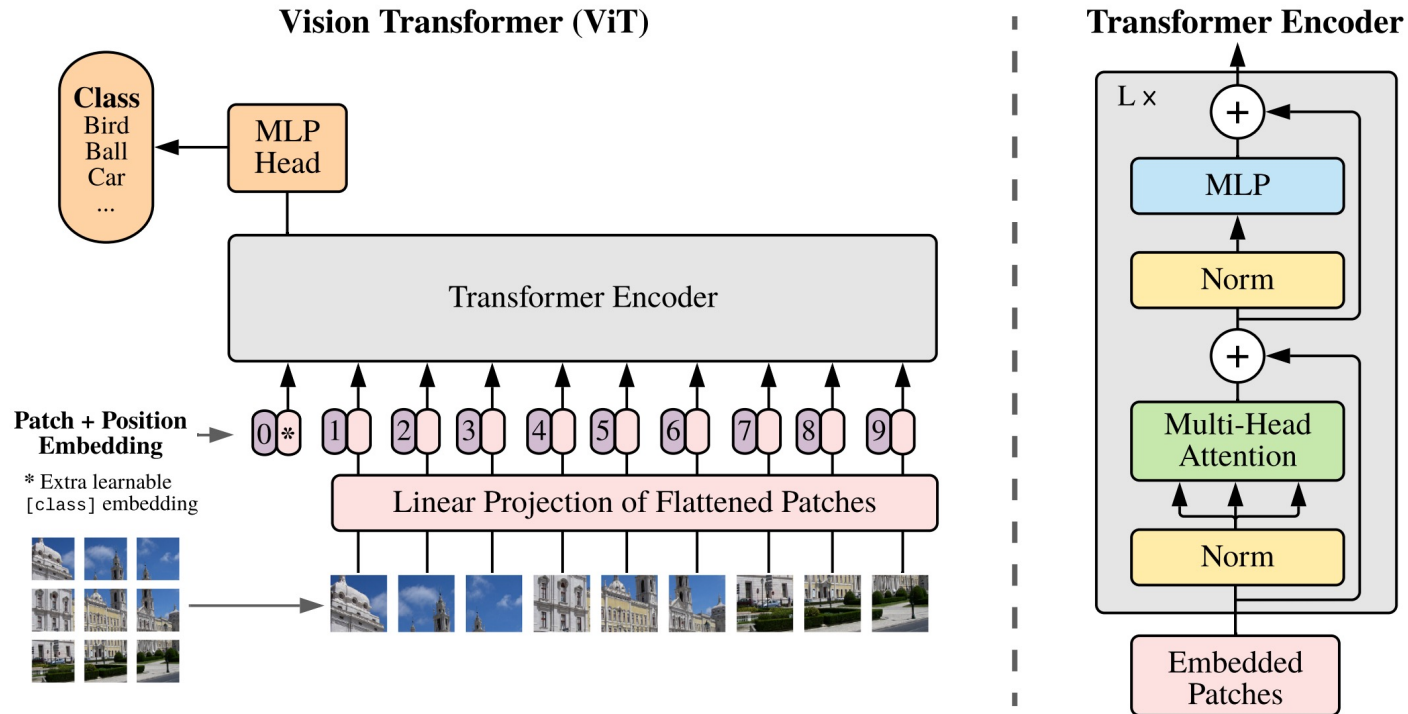
b) Convolutional neural networks



- **Not going into details.** Shift invariant architecture for image representation.
- Multiple layers of:
 - **Convolution + RELU**
 - **Convolution** layer puts the input image through a set of convolutional filters, each of which activates certain features in the image
 - **RELU**, a non-linear activation function - allows for effective training by mapping negative values to zero and maintaining positive values
 - **Pooling** - simplifies the output by performing a down-sampling

c) Vision transformers

Dosovitskiy et al. 2021



Create image tokens: Split image into image patches, map them into vectors and linearly project them to patch embeddings.

[CLS]: Add a learnable special token [CLS] embedding to the sequence

[Position embeddings]

Text Encoders

Key idea: Map a text input into a set of feature vectors $w = \{w_1, \dots, w_N\}$

e.g., BERT, RoBERTa

Multimodal Fusion

Key idea: A fusion encoder takes both visual features $v = \{v_1, \dots, v_M\}$ and text features $w = \{w_1, \dots, w_N\}$ as input, and maps them to **contextualized multimodal representations**: $\tilde{v} = \{\tilde{v}_1, \dots, \tilde{v}_M\}$ and $\tilde{w} = \{\tilde{w}_1, \dots, \tilde{w}_N\}$

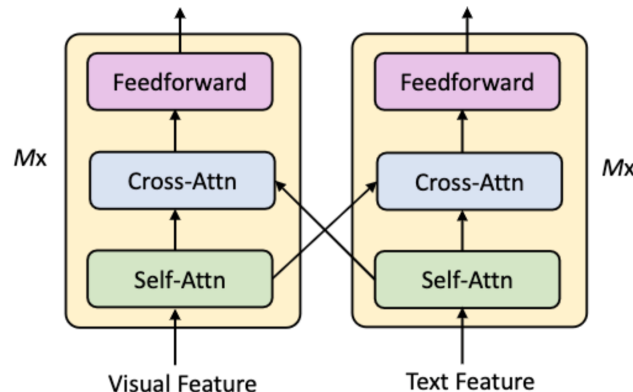
There are mainly two types of fusion modules:

a) Merged attention

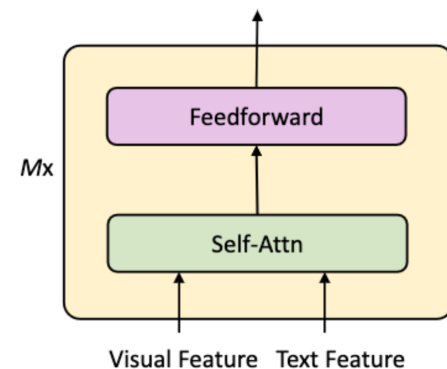
- **Concatenate** text and visual features and **feed them into** a single **Transformer** block.
- Used in many works, such as VisualBERT, Unicoder-VL, VL-BERT, UNITER, OSCAR, VinVL, ViLT, GIT

b) Co-attention

- Text and visual features fed into **different Transformer blocks**
- Within the transformer blocks, apply **cross-attention** to enable cross-modal interaction
- Used in LXMERT, ViLBERT, ERNIE-ViL, METER
- In recent large models, e.g. Flamingo (DeepMind), only image-to-text cross-attention is used



(a) Co-attention



(b) Merged attention

Multimodal Language Model Pretraining

Several pretraining objectives:

a) Masked Language Modelling

Given an image-text pair, **randomly mask out input words** with probability 15%, and predict the masked tokens based on their surrounding words **and** the paired image.

- In (**autoregressive**) models, only preceding context can be used to predict the masked token.

Some models adopt **efficient finetuning** (prefix tuning).

Multimodal Language Model Pretraining

b) Image Text Matching (ITM)

Given a batch of image-caption pairs, the key idea is to **identify which images and captions correspond to each other**.

A typical idea is to use **binary classification**.

- Append a special token (i.e., [CLS]) at the beginning of the input sentences and learn a global **cross-modal representation**.
- Feed the model with either a matched or mismatched image-caption pair with equal probability,
- Add a **classifier** on top of the [CLS] token to predict a binary label indicating whether the sampled image-caption pair is matched or not.

Multimodal Language Model Pretraining

c) Image Text Contrastive Learning (ITC)

A refinement of Image Text Matching (ITM).

Similar idea: Given a batch of N image-text pairs, the **idea** is again to predict the N matched pairs from all the N^2 possible image-text pairs. **However, the idea is to use contrastive learning.**

Let $\{v_i\}_{i=1}^N$ and $\{w_i\}_{i=1}^N$ be the normalized image and text vectors in a training batch. Then, we have:

$$s_{i,j}^{i2t} = v_i^\top w_j, \quad s_{i,j}^{t2i} = w_i^\top v_j,$$

The key idea is to encourage the matched image-text pairs to have higher similarity than unmatched image-text pairs.

Multimodal Language Model Pretraining

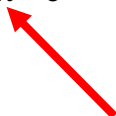
Contrastive Learning:

$$s_{i,j}^{i2t} = v_i^\top w_j, \quad s_{i,j}^{t2i} = w_i^\top v_j,$$
$$\mathcal{L}_{\text{ITC}}^{i2t}(\theta) = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(s_{i,i}^{i2t}/\sigma)}{\sum_{j=1}^N \exp(s_{i,j}^{i2t}/\sigma)},$$

Logistic regression

$$\mathcal{L}_{\text{ITC}}^{t2i}(\theta) = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(s_{i,i}^{t2i}/\sigma)}{\sum_{j=1}^N \exp(s_{i,j}^{t2i}/\sigma)},$$


Sometimes, a different loss function (like Noise Contrastive Estimator (NCE) loss is used):

$$\mathcal{L}_{\text{ITC}}^{i2t}(\theta) = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp\left(\frac{s_{i,i}^{i2t}}{\sigma}\right)}{\exp\left(\frac{s_{i,i}^{i2t}}{\sigma}\right) + \sum_{k=0}^K \exp\left(\frac{s_{i,k}^{i2t}}{\sigma}\right)}$$


K negative examples taken from remaining $N^2 - N$ unmatched pairs

Multimodal Language Model Pretraining

d) Masked image modelling

Reconstruct masked image patches/regions given the remaining image

(i) For object detection models:

- Replace visual features of the masked region by zeros, and **predict** the original region features via minimizing the **mean squared error loss**
- Some models **predict object labels** for masked regions instead of/before predicting image features

(ii) For end-to-end vision language pretrained models:

- Very much like the above; but can use **VQ-VAEs** (van den Oord et al., 2017) which are variational autoencoders which learn discrete latent representations.
- Discrete VQ tokens are extracted for the input patches, and the model is trained to reconstruct the discrete tokens.

Tasks for Multimodal Language Models

VQA & Visual Reasoning

Q: What is the dog holding with its paws?
A: Frisbee.

Text-to-Image Retrieval

Query: A dog is lying on the grass next to a frisbee.

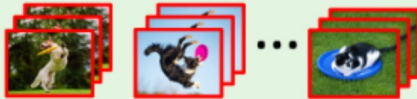
Negative Images



Text-to-Video Retrieval

Query: A dog is lying on the grass next to a frisbee, *while shaking its tail*.

Negative Videos



Video Question Answering

Q: Is the dog perfectly still?
A: No.

Image Captioning

Caption: A dog is lying on the grass next to a frisbee.

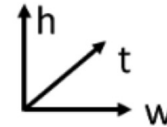


Image Classification

Labels: [dog, grass, frisbee]

Object Detection



dog, grass, frisbee

Segmentation



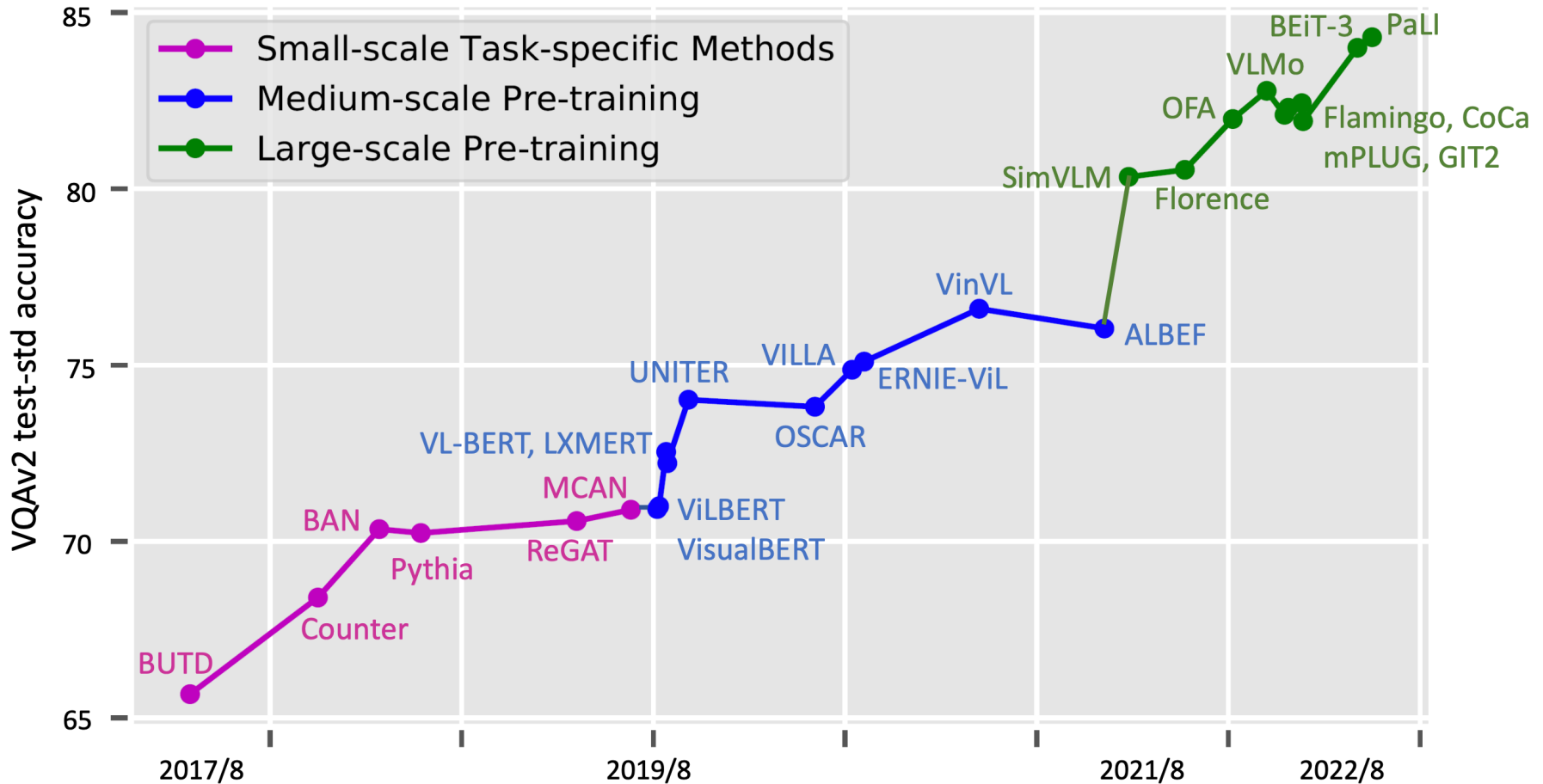
dog, grass, frisbee

Useful for image-text tasks ,

vision tasks as VL problems , and video-text tasks .

Progress in Multimodal Language Models

VQA task as a case study:



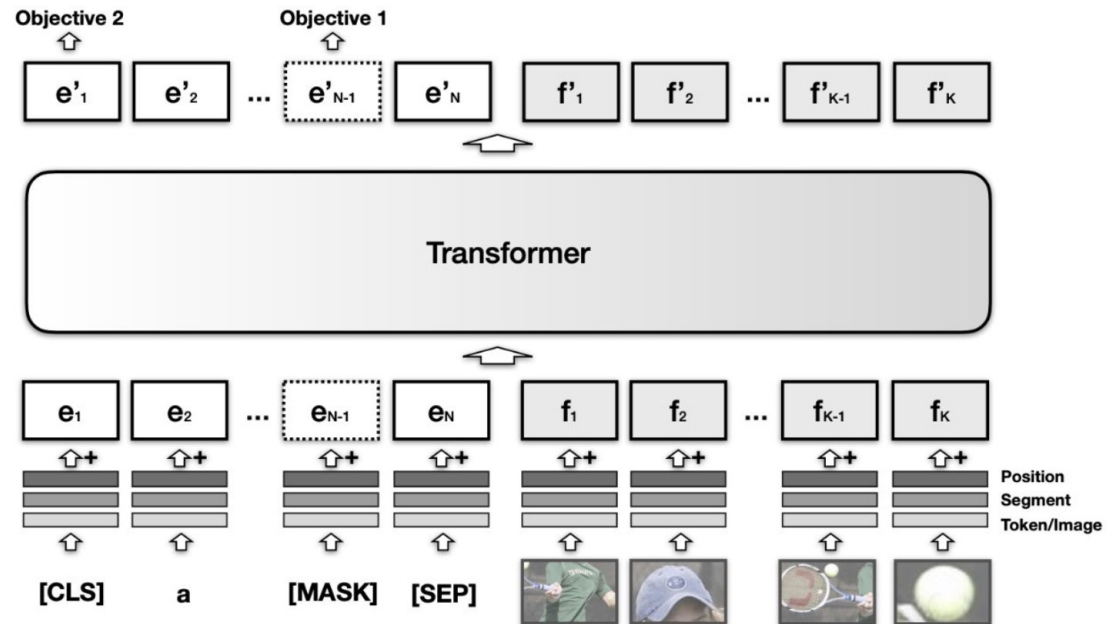
Transition from task-specific methods to large-scale pre-training.

Every time when there was a transition, we observe a big performance lift

Popular Models 1: VisualBERT



A person hits a ball with a tennis racket

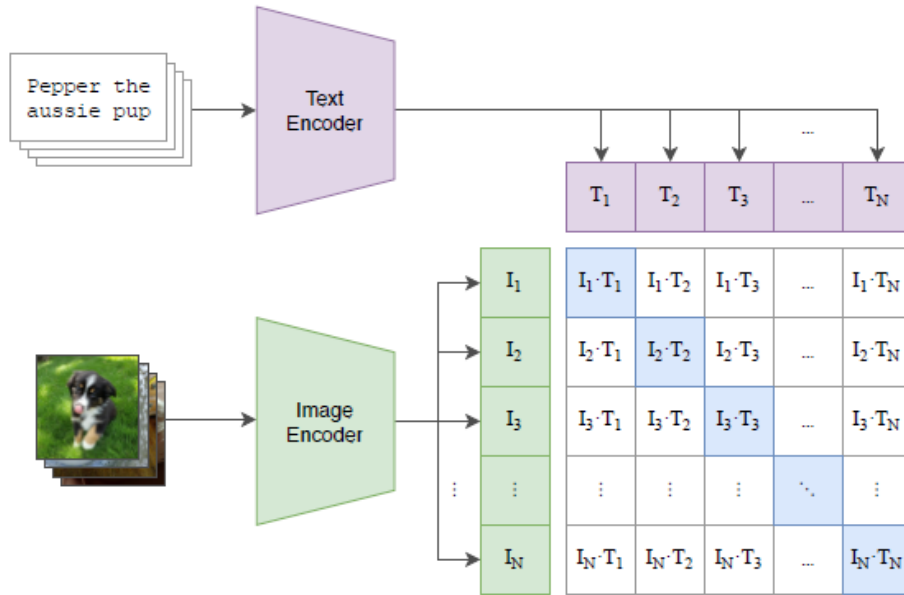


- Image regions and language are combined with a Transformer to allow self-attention to discover implicit alignments between language and vision.
- Pre-trained with a masked language modeling, and sentence-image prediction task (binary classification task for predicting which image is paired with which caption) on image-caption data

Popular Models 2: CLIP

Train time:
Contrastive pre-training

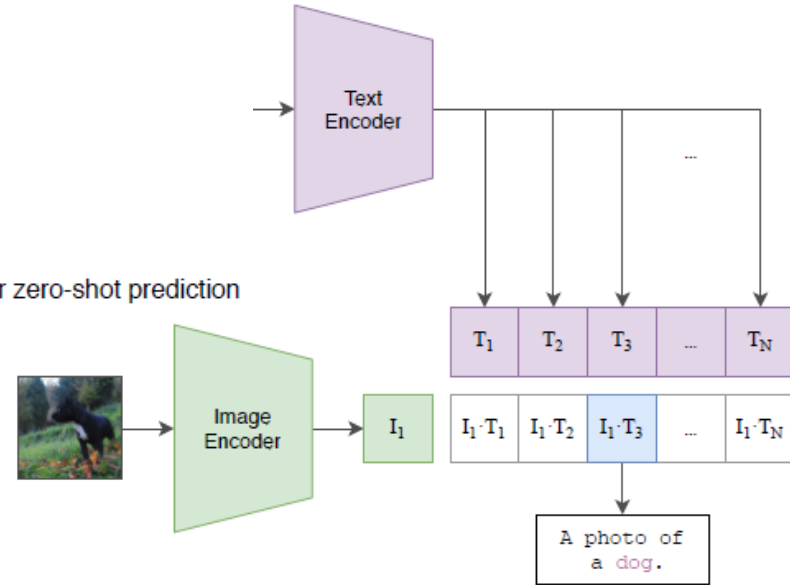
(1) Contrastive pre-training



Test time (a bit limited):

(1) Either just retrieve from existing captions (choose the caption with highest cosine similarity with test image)

(3) Use for zero-shot prediction



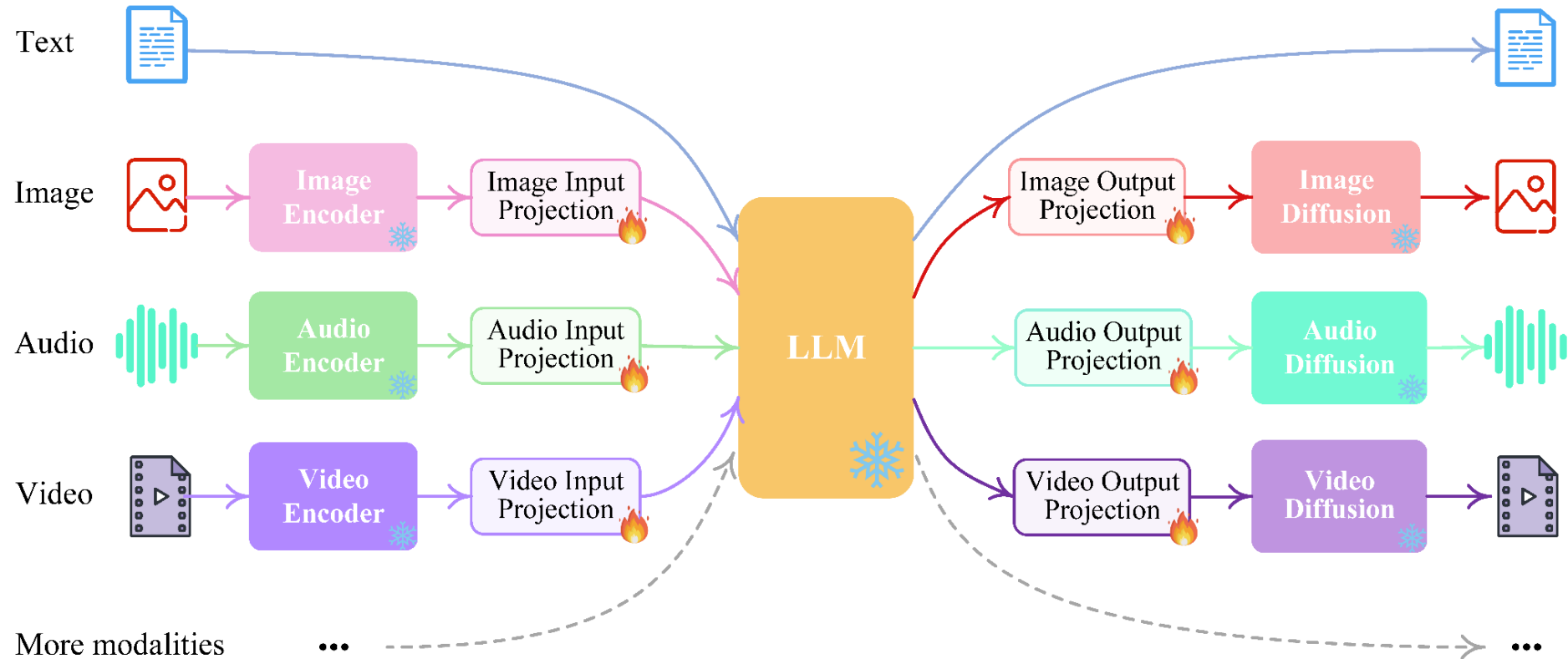
Learning objectives:

- Predict matched image given the caption
- Predict matched caption given image
- (Optionally) Contrastive loss:
maximize cosine similarity of the image and text embeddings of the N matched pairs in the batch while minimizing cosine similarity of embeddings of the $N^2 - N$ incorrect pairings

or (2) use a set of object classes to generate a large set of "synthetic" captions

Latest Models: GPT-4, AnyMaL, NexT-GPT

Several modalities as input or output:



Thanks!